
Lifecycle performance

CPU work and RTNL serialization

Fresh, pooled and bridge-free veth / VXLAN endpoints
Operation costs, resource limits and paths to parallelism

KERNEL MEASUREMENTS

October 5–8, 2026
VM1 and VM2 · 8 vCPUs each · Linux 6.12.95
Historical fixture scope · Concurrency 256

READING GUIDE

Summary p2 · Definitions p3 · CPU charts p4–6
RTNL charts p7–8 · Resource limits p9
Complete operation tables p10–13 · Next steps p14
Bridge-free columns use the same concurrency: 256
Dedicated encrypted endpoint lifecycle

Where performance stands

Fresh and pooled devices • C256 throughout • Bridge-free columns added October 8, 2026

Bridge-free forwarding substantially reduces pooled RTNL work while keeping dedicated per-edge encryption. The matched C256 comparison improves fresh-device turnover by 26–27% and pooled turnover by 79–81%.

C256	B cycles/s	B CPU ms	B RTNL ms	NB cycles/s	NB CPU ms	NB RTNL ms
Fresh · VM1	380	6.91	2.46	522	3.97	1.56
Fresh · VM2	382	6.85	2.42	527	3.92	1.55
Pooled · VM1	554	4.77	1.61	914	3.11	0.53
Pooled · VM2	566	4.60	1.57	953	2.95	0.50

B = historical bridge fixture (Oct 5–6); NB = bridge-free (Oct 8). Gains above use the matched Oct 8 bridged reruns, not ratios between these dates. CPU and RTNL overlap and must not be added. Rates count local activation/reset cycles, not requests or two-ended edges.

1 CPU overhead is real

Udev owned about 44% of sampled unfiltered host CPU. Early scoped filtering removes the event-worker storm and substantially improves fresh-device throughput.

2 Bridge-free pools remove the main lock hotspot

C256 pooled bridge-free RTNL is 0.530 / 0.503 ms/cycle, about 67% below the matched bridged reruns. Fixed redirects stay installed across leases; fresh keys and link-state changes remain.

3 Preserve the comparison scope

The same historical exact-tuple conntrack deletes and known-address reset are used. Extra gateway rules/routes are included. Product-wide table dumps, controller queues and persistence remain integration work; these are kernel rates.

Three clocks, two resource limits

An endpoint's lifecycle includes CPU work and RTNL waiting; CPU execution can overlap RTNL hold time.

CPU time: active execution

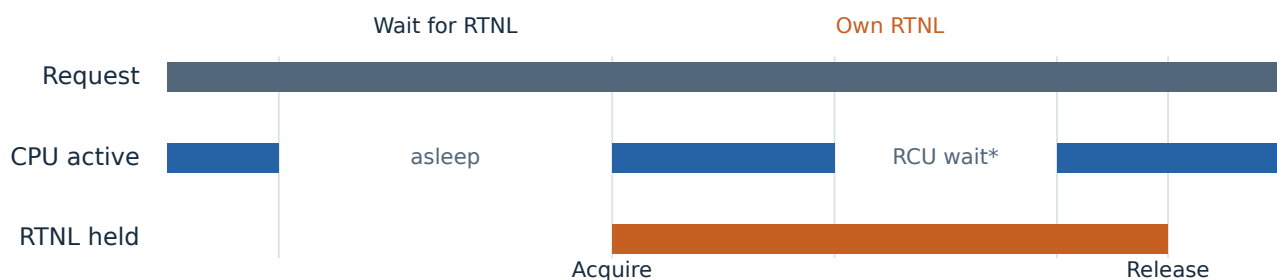
Sum of time executing on CPUs. Host CPU includes all processes and kernel activity across eight vCPUs; process CPU includes fixture threads and their kernel execution. Sleeping time is excluded.

RTNL hold time: a serialized resource

Elapsed time between acquiring and releasing the global networking lock. Sleeps, preemption and RCU waits while holding it still block other users. It overlaps CPU execution; it is not an extra CPU charge.

Request latency and lock wait: elapsed delay

A request may wait behind many concurrent callers before acquiring RTNL. Summed waits overlap across threads: 1,025 ms/cycle of aggregate acquisition wait in one trace does not mean one second of serial work per cycle.



*Illustrative sleep: caller uses no CPU; RTNL stays held. Timeline is not measured.

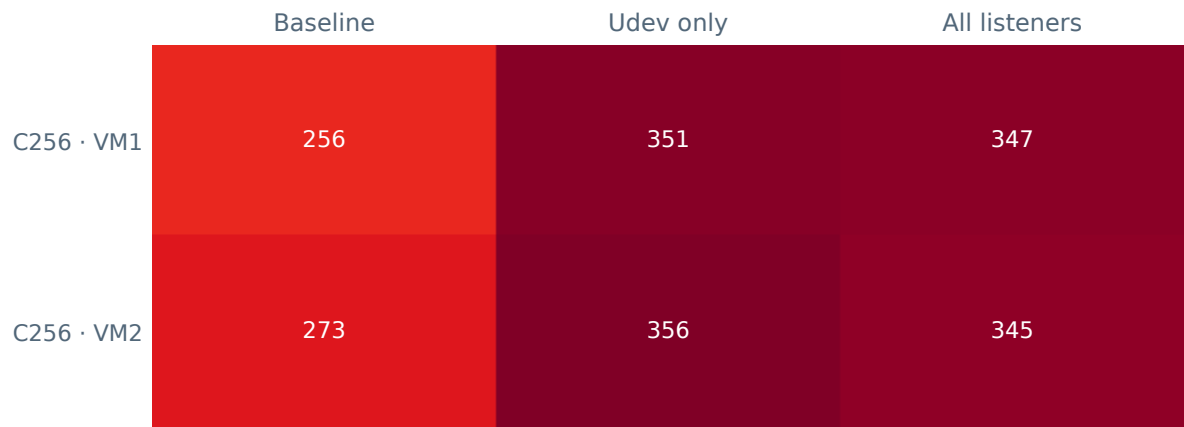
One cycle = one local endpoint setup/activation plus completed teardown/reset. A cross-host edge needs an endpoint on each host; per-host rates are not doubled to claim cross-host edge throughput. C is wave width; mixed trials can overlap C activations and C retirements.

Timed loops exclude initial pool preparation, final destruction, control RPCs, history/storage and application traffic. Empty fixture flow/neighbor tables make these lifecycle diagnostics with empty flow tables.

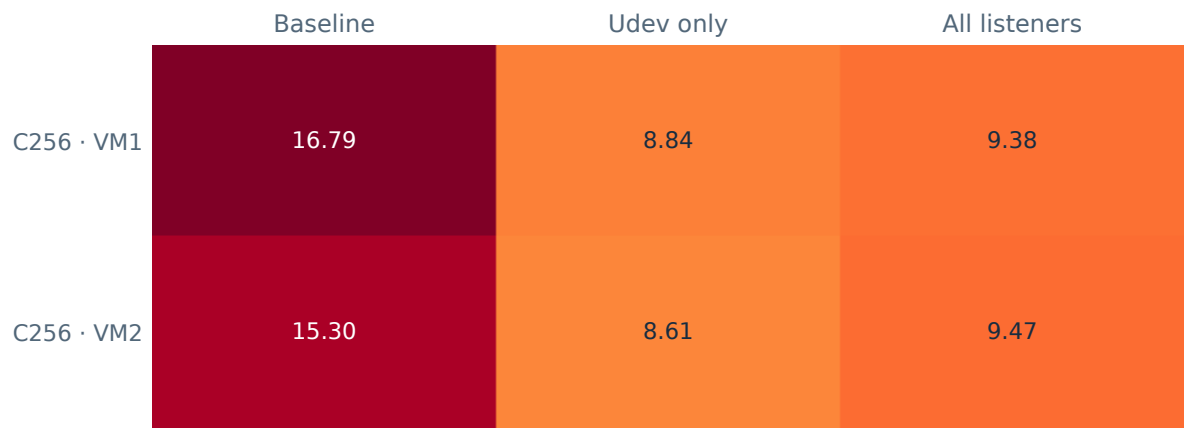
Filtering helps fresh devices most

Matched baseline → udev-only → all audited consumers. Two balanced 15-second trials per cell.

THROUGHPUT · complete local cycles/s



HOST CPU · ms per completed cycle

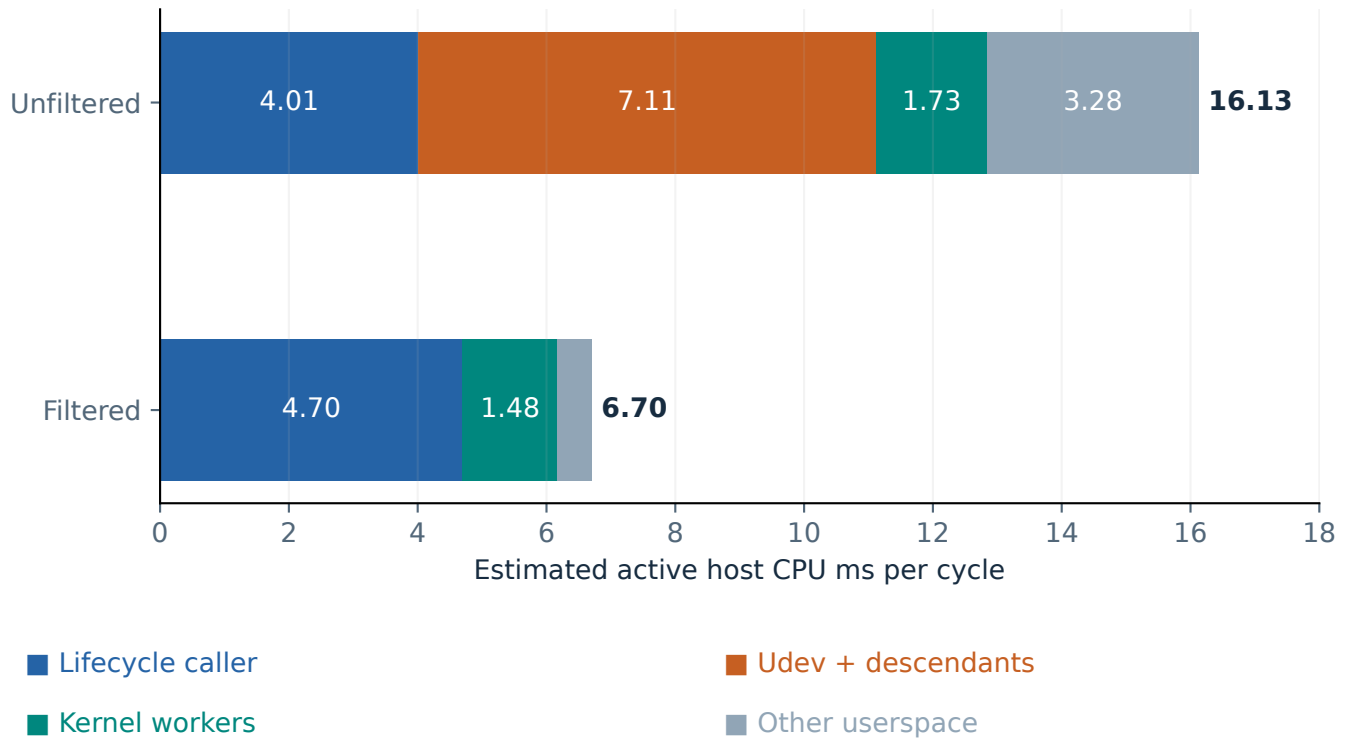


At C256, all-listener filtering improves throughput 26–35%. Udev provides most of that gain; adding the rtnetlink consumers does not improve throughput over udev-only in these matched trials.

Darker cells mean larger values within each heatmap, not necessarily better performance. Subsequent preparation-filtered runs reached 380–382/s at C256; those are separate runs, not matched speedup estimates. Baseline udev drain timed out after completed comparison trials on both hosts.

Where the host CPU goes

Execution-owner sampling exposes the notification cost that request timers cannot see.



Unfiltered: 3,328 cycles, 16.13 sampled CPU ms/cycle versus 16.56 from /proc/stat. Filtered: 6,144 cycles, 6.70 sampled versus 6.78 from /proc/stat. No lost samples; no udev samples after filtering is not a universal zero-CPU guarantee.

Why pooled filtering has little payoff

Pooled timed trials registered zero raw uevent drops: devices are not repeatedly registered. Link/address notifications still occur. Filtering changes throughput about 0-1% at C256.

Udev processes device hotplug events, rules and helpers. Avahi tracks interfaces/addresses for multicast DNS. Filtering only owned Nullnet virtual-device notifications preserves foreign, Docker and physical-device events. A udev rule runs after queueing; the socket filter prevents queueing/wakeup earlier.

CPU work by operation family

Calling-thread CPU, C256. Fresh and pooled device lifecycles, concurrency 256.

	Fresh B VM1	Fresh B VM2	Pool B VM1	Pool B VM2	Fresh NB VM1	Fresh NB VM2	Pool NB VM1	Pool NB VM2
Device creation	0.757	0.721	0.000	0.000	0.822	0.822	0.000	0.000
TC installation	0.375	0.367	0.000	0.000	0.684	0.697	0.000	0.000
IPsec install/remove	0.300	0.289	0.187	0.192	0.254	0.242	0.482	0.450
Addresses	0.274	0.269	0.376	0.368	0.120	0.113	0.330	0.316
Attach / enable	0.673	0.655	0.807	0.790	0.286	0.250	0.377	0.364
Disable links	0.312	0.279	0.453	0.415	0.219	0.191	0.427	0.401
Detach ports	0.000	0.000	0.408	0.399	0.000	0.000	0.000	0.000
GETLINK queries	0.958	0.943	0.410	0.407	0.659	0.642	0.239	0.233
Batch deletion	0.509	0.470	0.000	0.000	0.309	0.319	0.000	0.000
Conntrack deletes	0.239	0.222	0.072	0.079	0.180	0.194	0.199	0.187
Neighbor dump	0.145	0.142	0.124	0.106	0.000	0.000	0.189	0.179
Peer routes	0.000	0.000	0.000	0.000	0.109	0.098	0.313	0.291
Total	4.542	4.356	2.837	2.755	3.642	3.569	2.556	2.421

Operation family	What it does	Why needed
Device creation	Create the veth pair and VXLAN interface.	Build this dedicated endpoint; retained devices skip it.
TC installation	Add packet-hook attachment points and TC filters.	Attach the packet steering/isolation programs; pools keep them.
IPsec install/remove	Add/remove encryption states and the outbound policy.	Encrypt the edge; remove old keys/policy on retirement.
Addresses	Add/remove endpoint IP addresses on interfaces.	Give the endpoint IP connectivity; clear addresses on reset.
Attach / enable	Join interfaces to the bridge and set interfaces UP.	Connect the forwarding path and make interfaces operational.
Disable links	Set interfaces DOWN while keeping the devices.	Quiesce the old endpoint during teardown/reset.
Detach ports	Remove retained interfaces from bridge membership.	Disconnect retained bridge ports in the current reset design.
GETLINK queries	Read interface indices and check interface state.	Resolve device IDs and synchronize readiness; verify reset.
Batch deletion	Delete fresh veth/VXLAN devices in one batch.	Reclaim fresh devices; retained devices skip deletion.
Conntrack deletes	Remove scoped connection-tracking and NAT entries.	Clear old flow/NAT state before reuse; tables are empty here.
Neighbor dump	List IP-to-link-layer address mappings for cleanup.	Find stale ARP/NDP entries to clear; tables are empty here.
Peer routes	Route host traffic to remote endpoint/gateway IPs.	Make remote endpoint/gateway IPs reachable from the host.

B = original bridge measurements; NB = Oct 8 bridge-free. Fresh B CPU is unfiltered; the other columns are filtered. Different trials; full rows and scope are on p10/p12. External consumers and sleeping lock waits are excluded.

GETLINK: bridge fresh has six setup checks + one reset check; bridge-free fresh has five setup checks. Pools have four (B) / three (NB) readiness checks.

Lifecycle control only; packet encryption and phase-barrier experiments are excluded.

Sources: lifecycle-cpu concurrent_request_cpu_ms; retained cpu/256/rows. Calling-thread active CPU.

RTNL hold time by operation family

Fine request-correlated RTNL holds with filtering enabled, concurrency 256.

	Fresh B VM1	Fresh B VM2	Pool B VM1	Pool B VM2	Fresh NB VM1	Fresh NB VM2	Pool NB VM1	Pool NB VM2
Device creation	0.583	0.585	0.000	0.000	0.553	0.548	0.000	0.000
TC installation	0.068	0.065	0.000	0.000	0.140	0.135	0.000	0.000
Attach / enable	0.389	0.380	0.392	0.387	0.083	0.078	0.088	0.084
Disable links	0.550	0.537	0.434	0.403	0.240	0.245	0.305	0.289
Detach ports	0.000	0.000	0.641	0.636	0.000	0.000	0.000	0.000
Batch deletion	0.711	0.699	0.000	0.000	0.449	0.454	0.000	0.000
Addresses	0.072	0.071	0.086	0.085	0.043	0.041	0.081	0.078
GETLINK queries	0.082	0.081	0.056	0.055	0.035	0.033	0.025	0.024
Peer routes	0.000	0.000	0.000	0.000	0.016	0.014	0.031	0.029
Total	2.455	2.417	1.609	1.565	1.558	1.549	0.530	0.503

Operation family	What it does	Why needed
Device creation	Create the veth pair and VXLAN interface.	Build this dedicated endpoint; retained devices skip it.
TC installation	Add packet-hook attachment points and TC filters.	Attach the packet steering/isolation programs; pools keep them.
Attach / enable	Join interfaces to the bridge and set interfaces UP.	Connect the forwarding path and make interfaces operational.
Disable links	Set interfaces DOWN while keeping the devices.	Quiesce the old endpoint during teardown/reset.
Detach ports	Remove retained interfaces from bridge membership.	Disconnect retained bridge ports in the current reset design.
Batch deletion	Delete fresh veth/VXLAN devices in one batch.	Reclaim fresh devices; retained devices skip deletion.
Addresses	Add/remove endpoint IP addresses on interfaces.	Give the endpoint IP connectivity; clear addresses on reset.
GETLINK queries	Read interface indices and check interface state.	Resolve device IDs and synchronize readiness; verify reset.
Peer routes	Route host traffic to remote endpoint/gateway IPs.	Make remote endpoint/gateway IPs reachable from the host.

B = Oct 6 bridge traces; NB = Oct 8 bridge-free traces. NB RTNL totals: fresh 1.558 / 1.549 ms; pooled 0.530 / 0.503 ms. Lock-held elapsed time includes sleeps/preemption; it is not active CPU.

Fresh: creation, deletion and VXLAN/bridge disable account for about 70% of RTNL; deletion holds 182 / 179 ms per 256-endpoint batch.

Pooled: link changes account for about 91% of RTNL; detaching ports accounts for about 40%. No RTNL was observed for IPsec, conntrack or neighbor requests; those still consume CPU.

Sources: Oct 6 fine-lock summary; retained locks/256. All units: elapsed RTNL ms/cycle.

Pooled RTNL without bridges

Filtered C256 • Original bridge results plus bridge-free operation columns.

Heatmap: darker = larger ms/cycle; shared VM scale per page; subtotals/totals excluded.

Operation	B VM1	B VM2	NB VM1	NB VM2
Detach outer veth	0.4271	0.4225	0.0000	0.0000
Detach VXLAN	0.2143	0.2137	0.0000	0.0000
Disable container peer	0.1716	0.1410	0.1117	0.0991
Attach / enable outer	0.1662	0.1627	0.0348	0.0329
Disable VXLAN	0.1361	0.1357	0.0997	0.1004
Disable bridge	0.1260	0.1260	0.0000	0.0000
Attach / enable VXLAN	0.1168	0.1156	0.0215	0.0201
Enable bridge	0.0716	0.0715	0.0000	0.0000
Disable outer veth	0.0000	0.0000	0.0936	0.0891
Add / remove peer routes	0.0000	0.0000	0.0307	0.0290
Total fixture RTNL	1.6090	1.5653	0.5299	0.5035

Bridge membership disappears; link and route work remains

Fixed TC redirects replace bridge membership. Pooled activation and reset still enable/disable the dedicated devices, change addresses and peer routes, replace encryption state and clear stale state. Redirect installation is outside each lease.

B = original Oct 6 bridge traces; NB = Oct 8 bridge-free traces. The new matched bridged reruns held RTNL 1.582 / 1.540 ms/cycle, versus 0.530 / 0.503 without bridges: a 66.5–67.3% reduction. Comparisons use the same historical cleanup scope.

Rows above highlight the former hotspot and its replacement operations; p13 accounts for every measured operation. CPU and RTNL are separate overlapping budgets. Initial preparation and final pool destruction remain outside lease timing.

Bridge-free budgets and achieved rates

Filtered C256 • Eight CPUs • Separate profiler-free and RTNL trials.

Bridge-free	Observed /s	CPU ms	RTNL ms	CPU cap /s	RTNL cap /s
Fresh VM1	522	3.968	1.558	2016	642
Fresh VM2	527	3.922	1.549	2040	646
Pooled VM1	914	3.113	0.530	2570	1887
Pooled VM2	953	2.951	0.503	2711	1986

Removing bridges lifts the serialized-work ceiling

Fresh bridge-free cycles still create/delete dedicated devices and hold RTNL about 1.55 ms/cycle. Pools retain those devices and hold about 0.50–0.53 ms/cycle, leaving link state and route changes as necessary work.

Observed matched gains are 26–27% fresh and 79–81% pooled. Sixty-second C256 confirmations reach 507 / 509 cycles/s fresh and 875 / 881 pooled. The short repeated runs above reach 522 / 527 and 914 / 953; sustained confirmation does not establish 1,000/s.

A resource ceiling is not a throughput promise

CPU cap = 8,000 / host CPU ms per cycle; RTNL cap = 1,000 / lock-held ms per cycle. These assume unchanged work and ideal utilization. Admission, scheduling, other lock users and completed-reset drains can keep throughput below either ceiling.

The same historical reset scope is retained. Product-wide conntrack/address dumps, controller work and other integration costs are separate; these kernel rates are not product request or bulk packet throughput.

Fresh-device CPU: complete attribution

C256 • Bridge: unfiltered Oct 5 • Bridge-free: filtered Oct 8 • Calling-thread CPU ms/cycle

Heatmap: darker = larger ms/cycle; shared VM scale per page; subtotals/totals excluded.

Operation / combined calls	B VM1	B VM2	NB VM1	NB VM2
SETUP · Veth creation	0.5208	0.4940	0.5258	0.5193
VXLAN creation	0.2367	0.2268	0.2958	0.3027
TC qdisc(s) + filters	0.3750	0.3668	0.6841	0.6974
Two IPsec states + outbound policy	0.1997	0.1991	0.1853	0.1786
Two addresses	0.2079	0.2085	0.1204	0.1127
Two attach + enable requests	0.4202	0.4030	0.0000	0.0000
Bridge + container peer enable	0.2530	0.2517	0.0739	0.0699
Setup GETLINK lookups/checks	0.9024	0.8886	0.6593	0.6417
Enable outer veth without bridge	0.0000	0.0000	0.0974	0.0747
Enable VXLAN without bridge	0.0000	0.0000	0.0669	0.0615
Enable transport again	0.0000	0.0000	0.0478	0.0441
Add two host peer routes	0.0000	0.0000	0.1087	0.0984
RESET · Two disable requests	0.3119	0.2792	0.2194	0.1912
Remove IPsec states + policy	0.1000	0.0895	0.0688	0.0636
Acknowledged batch device deletion	0.5089	0.4696	0.3088	0.3191
Four scoped conntrack deletions	0.2391	0.2221	0.1800	0.1943
Remove host gateway address	0.0662	0.0605	0.0000	0.0000
Bridge GETLINK check	0.0556	0.0545	0.0000	0.0000
Synchronous request subtotal	4.3973	4.2138	3.6425	3.5690
Neighbor dump · separate from requests	0.1445	0.1424	0.0000	0.0000
Total measured operation CPU	4.5418	4.3562	3.6425	3.5690

Original bridge CPU repeats covered 7,936 / 8,448 cycles. Bridge-free counts and integrity are in the Oct 8 JSON. Calling-thread CPU excludes sleeps and other execution owners. Original neighbor dumping is separate from request timers; overlapping inclusive timers are not added.

B = original unfiltered Oct 5 bridge CPU; NB = filtered Oct 8 bridge-free CPU. Both use C256 and the historical reset scope. TC rows combine one qdisc/three filters (B) and two qdiscs/eight filters (NB). Host gateway addressing remains; bridge-free host routes and direct link enables are explicit.

Fresh-device RTNL: complete attribution

Filtered C256 • Bridge: Oct 6 • Bridge-free: Oct 8 • RTNL hold ms per local cycle

Heatmap: darker = larger ms/cycle; shared VM scale per page; subtotals/totals excluded.

Operation / combined calls	B VM1	B VM2	NB VM1	NB VM2
SETUP · Create veth pair	0.4654	0.4643	0.4461	0.4405
Look up container peer	0.0084	0.0086	0.0078	0.0077
Create VXLAN	0.1172	0.1207	0.1070	0.1080
Look up VXLAN index	0.0076	0.0081	0.0068	0.0070
TC qdisc(s)	0.0173	0.0175	0.0356	0.0357
TC filters	0.0511	0.0474	0.1043	0.0993
Two IPsec states + policy	0.0000	0.0000	0.0000	0.0000
Add container address	0.0268	0.0265	0.0205	0.0196
Add host gateway address	0.0287	0.0281	0.0223	0.0212
Attach outer veth + enable	0.1753	0.1715	0.0000	0.0000
Attach VXLAN + enable	0.1227	0.1196	0.0000	0.0000
Enable bridge	0.0582	0.0565	0.0000	0.0000
Enable container peer	0.0331	0.0322	0.0275	0.0269
Readiness: VXLAN	0.0109	0.0104	0.0069	0.0063
Readiness: outer veth	0.0115	0.0114	0.0070	0.0063
Readiness: bridge	0.0238	0.0228	0.0000	0.0000
Readiness: container peer	0.0073	0.0071	0.0065	0.0058
Enable outer veth without bridge	0.0000	0.0000	0.0345	0.0322
Enable VXLAN without bridge	0.0000	0.0000	0.0189	0.0177
Enable transport again	0.0000	0.0000	0.0018	0.0016
Add two host peer routes	0.0000	0.0000	0.0161	0.0144
RESET · Disable VXLAN	0.2341	0.2320	0.2400	0.2449
Remove IPsec states + policy	0.0000	0.0000	0.0000	0.0000
Batch-delete VXLAN + veths	0.7110	0.6986	0.4489	0.4537
Four scoped conntrack deletes	0.0000	0.0000	0.0000	0.0000
Disable bridge	0.3156	0.3050	0.0000	0.0000
Dump bridge neighbors	0.0000	0.0000	0.0000	0.0000
Remove host gateway address	0.0163	0.0160	0.0000	0.0000
Verify bridge GETLINK	0.0129	0.0129	0.0000	0.0000
Setup subtotal	1.1652	1.1528	0.8695	0.8501
Teardown subtotal	1.2899	1.2645	0.6889	0.6986
Total fixture RTNL hold time	2.4551	2.4172	1.5584	1.5487

Zero means no RTNL observed, not zero CPU. Batch deletion is divided by 256 endpoints. Other actors add 0.106 / 0.101 ms/cycle, excluded here. Fresh CPU attribution is on p10.

B = original Oct 6 bridge traces; NB = Oct 8 bridge-free traces. Same C256 historical lifecycle scope, separate trials. Zero can mean an absent operation or no observed RTNL. Batch deletion implicitly removes fresh bridge-free gateway addresses and peer routes.

Pooled CPU: complete attribution

Filtered C256. Milliseconds of calling-thread CPU per complete local cycle.

Heatmap: darker = larger ms/cycle; shared VM scale per page; subtotals/totals excluded.

Operation / combined calls	B VM1	B VM2	NB VM1	NB VM2
ACTIVATE · Two IPsec states	0.0828	0.0844	0.2029	0.1820
Outbound IPsec policy	0.0373	0.0367	0.0910	0.0828
Add container address	0.0720	0.0716	0.0943	0.0893
Enable container peer	0.0853	0.0855	0.1092	0.1036
Add host gateway address	0.0909	0.0870	0.0965	0.0920
Attach outer veth + enable	0.2374	0.2296	0.0000	0.0000
Attach VXLAN + enable	0.2169	0.2117	0.0000	0.0000
Enable bridge	0.1599	0.1582	0.0000	0.0000
Enable transport again	0.1074	0.1052	0.0566	0.0567
Readiness: VXLAN	0.1171	0.1172	0.0807	0.0789
Readiness: outer veth	0.1057	0.1024	0.0791	0.0773
Readiness: bridge	0.1054	0.1052	0.0000	0.0000
Readiness: container peer	0.0816	0.0818	0.0795	0.0773
Enable outer veth without bridge	0.0000	0.0000	0.1134	0.1095
Enable VXLAN without bridge	0.0000	0.0000	0.0977	0.0942
Add two host peer routes	0.0000	0.0000	0.1489	0.1431
RESET · Disable VXLAN	0.1204	0.1165	0.1297	0.1215
Disable bridge	0.1364	0.1323	0.0000	0.0000
Detach outer veth	0.1956	0.1894	0.0000	0.0000
Detach VXLAN	0.2125	0.2094	0.0000	0.0000
Disable container peer	0.1962	0.1658	0.1551	0.1411
Remove outbound IPsec policy	0.0256	0.0267	0.0659	0.0636
Remove two IPsec states	0.0412	0.0445	0.1224	0.1214
Four exact conntrack deletes	0.0720	0.0790	0.1986	0.1870
Dump container neighbors	0.0276	0.0277	0.0577	0.0553
Dump bridge neighbors	0.0967	0.0784	0.0000	0.0000
Remove container address	0.0974	0.0965	0.0638	0.0603
Remove host gateway address	0.1158	0.1130	0.0756	0.0742
Disable outer veth without bridge	0.0000	0.0000	0.1421	0.1380
Remove two host peer routes	0.0000	0.0000	0.1640	0.1477
Dump root veth / VXLAN neighbors	0.0000	0.0000	0.1308	0.1240
Activation subtotal	1.4996	1.4764	1.2499	1.1866
Reset subtotal	1.3373	1.2791	1.3057	1.2340
Total measured CPU	2.8369	2.7555	2.5556	2.4206

Original bridge repeats covered 13,824 / 14,080 cycles. CPU excludes sleeps, key derivation and argument/label construction before requests. Original instrumented process CPU is 3.25 / 3.18 ms/cycle. Bridge-free counts remain in the Oct 8 JSON.

B = Oct 6 bridge CPU; NB = Oct 8 bridge-free CPU. Filtered C256, same historical reset. Fixed pooled redirects are installed during preparation, not per lease. Gateway address, peer routes, direct link state and both root neighbor dumps are included.

Pooled RTNL: complete attribution

Filtered C256. Milliseconds holding RTNL per complete local cycle.

Heatmap: darker = larger ms/cycle; shared VM scale per page; subtotals/totals excluded.

Operation / combined calls	B VM1	B VM2	NB VM1	NB VM2
ACTIVATE · Two IPsec states	0.0000	0.0000	0.0000	0.0000
Outbound IPsec policy	0.0000	0.0000	0.0000	0.0000
Add container address	0.0235	0.0230	0.0221	0.0215
Enable container peer	0.0344	0.0344	0.0296	0.0287
Add host gateway address	0.0313	0.0302	0.0267	0.0253
Attach outer veth + enable	0.1662	0.1627	0.0000	0.0000
Attach VXLAN + enable	0.1168	0.1156	0.0000	0.0000
Enable bridge	0.0716	0.0715	0.0000	0.0000
Enable transport again	0.0027	0.0026	0.0023	0.0022
Readiness: VXLAN	0.0117	0.0116	0.0082	0.0080
Readiness: outer veth	0.0112	0.0107	0.0084	0.0079
Readiness: bridge	0.0252	0.0246	0.0000	0.0000
Readiness: container peer	0.0079	0.0077	0.0081	0.0078
Enable outer veth without bridge	0.0000	0.0000	0.0348	0.0329
Enable VXLAN without bridge	0.0000	0.0000	0.0215	0.0201
Add two host peer routes	0.0000	0.0000	0.0193	0.0178
RESET · Disable VXLAN	0.1361	0.1357	0.0997	0.1004
Disable bridge	0.1260	0.1260	0.0000	0.0000
Detach outer veth	0.4271	0.4225	0.0000	0.0000
Detach VXLAN	0.2143	0.2137	0.0000	0.0000
Disable container peer	0.1716	0.1410	0.1117	0.0991
Remove outbound IPsec policy	0.0000	0.0000	0.0000	0.0000
Remove two IPsec states	0.0000	0.0000	0.0000	0.0000
Four exact conntrack deletes	0.0000	0.0000	0.0000	0.0000
Dump container neighbors	0.0000	0.0000	0.0000	0.0000
Dump bridge neighbors	0.0000	0.0000	0.0000	0.0000
Remove container address	0.0144	0.0146	0.0144	0.0139
Remove host gateway address	0.0170	0.0170	0.0181	0.0176
Disable outer veth without bridge	0.0000	0.0000	0.0936	0.0891
Remove two host peer routes	0.0000	0.0000	0.0114	0.0112
Dump root veth / VXLAN neighbors	0.0000	0.0000	0.0000	0.0000
Activation subtotal	0.5024	0.4948	0.1810	0.1722
Reset subtotal	1.1066	1.0705	0.3488	0.3313
Total measured RTNL hold time	1.6090	1.5653	0.5299	0.5035

Original bridge traces covered 12,288 cycles per host. Bridge-free counts and integrity remain in the Oct 8 JSON. Holds include sleeps/preemption while RTNL stays locked; acquisition waits are excluded. Zero means no observed RTNL, not zero CPU.

B = Oct 6 bridge traces; NB = Oct 8 bridge-free traces. Filtered C256, same historical reset. Direct enables/disables and route updates remain after bridge membership disappears. CPU and RTNL overlap and must not be added.